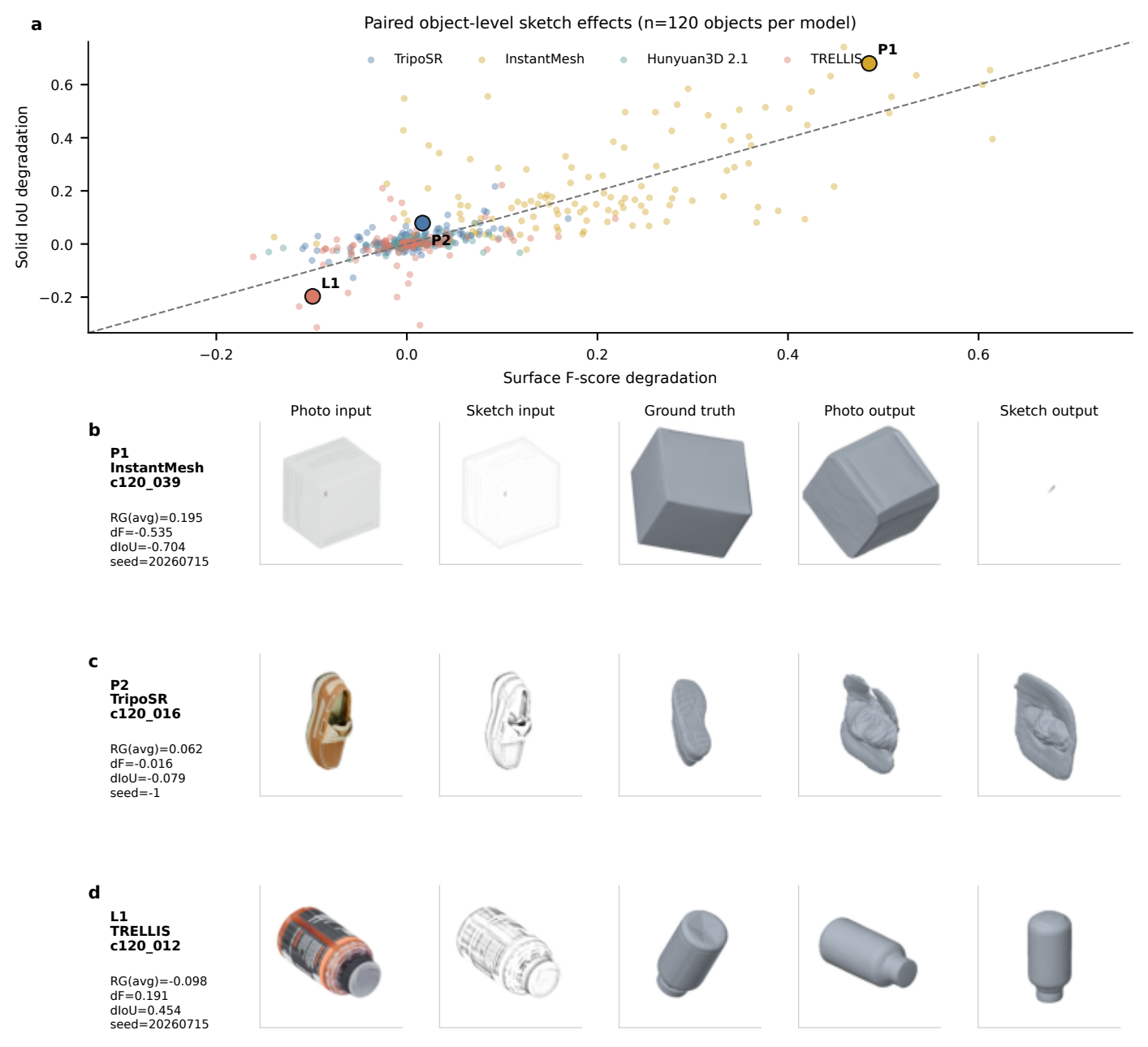

a
Paired object-level sketch effects (n=120 objects per model)
TripoSR
InstantMesh
Hunyuan3D 2.1
TRELLIS
P1
P2
L1
Solid IoU degradation
Surface F-score degradation
0.6
0.4
0.2
0.0
−0.2
−0.2
0.0
0.2
0.4
0.6
b
Photo input
Sketch input
Ground truth
Photo output
Sketch output
P1
InstantMesh
c120_039
RG(avg)=0.195
dF=-0.535
dIoU=-0.704
seed=20260715
c
P2
TripoSR
c120_016
RG(avg)=0.062
dF=-0.016
dIoU=-0.079
seed=-1
d
L1
TRELLIS
c120_012
RG(avg)=-0.098
dF=0.191
dIoU=0.454
seed=20260715